

CARROLL, IA, USA

WMO: 725468

Lat: 42.046N

Lon: 94.789W

Elev: 1204

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-1.5	-14.7	2.6	-4.4	-9.0	3.5	-0.2	34.1	16.8	31.5	21.3	8.7	290	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	92.6	76.6	90.2	75.9	87.7	74.2	80.4	88.7	78.4	86.4	76.4	84.4	13.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	155.3	87.0	75.4	139.3	84.0	73.2	129.1	81.8	45.2	88.2	43.0	86.1	40.6	85.5	95.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.9	25.9	23.3	-14.6	97.6	4.5	4.0	-17.8	100.5	-20.4	102.8	-22.9	105.1	-26.1	108.0		
			-14.3	84.5	4.9	4.2	-17.8	87.6	-20.7	90.1	-23.4	92.4	-27.0	95.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	21.7	24.2	37.0	49.9	61.2	71.6	74.0	71.5	64.8	52.2	38.1	25.4
	DBStd	21.24	12.50	12.95	12.34	9.98	8.98	6.64	5.67	5.60	8.46	10.18	10.82	11.74
	HDD50	3412	878	724	436	122	12	0	0	0	4	94	378	764
	HDD65	6586	1342	1143	871	460	183	17	5	10	106	414	806	1229
	CDD50	3206	0	1	31	120	360	648	745	666	448	164	22	1
	CDD65	902	0	0	1	7	65	215	285	210	101	18	0	0
	CDH74	9279	0	0	15	163	741	2237	2869	1923	1092	233	6	0
	CDH80	3405	0	0	1	45	258	890	1109	644	388	70	0	0
(14) Wind	WSAvg	10.2	11.9	11.6	11.9	12.9	12.0	9.6	6.9	6.1	7.7	10.1	11.4	11.0
(15) Precipitation	PrecAvg	33.0	0.8	1.0	2.1	3.5	4.6	5.2	4.2	4.2	2.5	2.0	1.4	1.0
	PrecMax	41.9	1.7	2.3	5.4	7.9	7.8	10.7	11.7	10.6	5.5	4.5	3.3	2.1
	PrecMin	24.4	0.1	0.2	0.0	1.0	1.5	1.2	1.3	0.6	0.6	0.2	0.0	0.0
	PrecStd	5.0	0.4	0.6	1.4	1.7	2.0	2.3	2.6	2.8	1.4	1.1	0.9	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	61.3	75.4	85.9	92.3	95.3	95.3	94.6	93.1	87.8	73.4	58.9
		MCWB	44.7	51.7	58.8	63.2	72.7	75.7	80.3	78.9	73.6	67.3	60.0	52.7
	2%	DB	46.3	53.9	68.4	79.0	86.3	91.3	91.2	90.3	89.5	80.9	66.1	50.4
		MCWB	40.5	47.7	56.3	61.7	69.0	74.8	78.6	78.3	72.6	64.5	56.1	45.3
	5%	DB	41.3	46.0	62.5	72.8	81.8	88.4	89.8	86.2	84.2	74.5	60.8	45.4
		MCWB	36.7	41.6	53.6	57.2	64.9	72.5	78.1	76.4	71.0	61.4	51.4	40.4
	10%	DB	36.9	40.8	55.1	68.0	77.3	85.7	86.2	82.4	81.1	69.9	54.8	40.7
		MCWB	34.2	37.3	47.9	55.0	63.0	70.9	76.2	73.8	68.3	59.0	47.2	37.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	53.7	63.4	66.4	75.9	80.5	83.8	82.8	78.3	73.4	61.9	54.2
		MCDB	52.4	60.6	73.8	79.9	87.6	90.1	92.6	91.3	87.4	82.5	70.5	58.4
	2%	WB	41.1	47.4	57.9	63.1	71.6	78.0	81.1	79.7	75.2	67.4	57.3	45.5
		MCDB	46.6	52.4	64.7	74.4	82.2	87.5	89.0	88.2	84.2	76.1	64.7	48.9
	5%	WB	37.3	41.8	53.5	59.6	68.2	75.4	79.1	77.4	73.1	63.9	52.3	40.7
		MCDB	41.2	46.7	61.8	71.0	77.8	84.7	86.7	84.7	81.3	70.8	59.0	45.0
	10%	WB	34.4	37.6	48.4	56.3	65.8	73.3	76.9	75.1	70.7	59.7	48.2	37.1
		MCDB	37.0	40.7	56.4	66.0	75.1	82.7	84.5	81.9	78.4	68.1	55.0	40.9
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.0	19.2	23.0	21.9	21.0	20.5	20.8	24.7	23.0	19.8	15.9
		MCDBR	22.4	23.1	29.1	33.2	28.3	24.5	22.8	23.8	27.6	30.3	28.5	22.8
	5% WB	MCWBR	17.4	17.7	18.6	19.1	14.2	12.0	13.6	15.1	15.0	16.9	18.2	17.9
		MCWBR	20.5	22.8	26.9	28.9	24.0	21.2	20.3	21.3	23.1	24.7	25.8	21.4
(40) Clear-Sky Solar Irradiance	taub		0.282	0.299	0.320	0.373	0.398	0.415	0.427	0.414	0.381	0.336	0.295	0.280
		taud	2.421	2.386	2.419	2.293	2.295	2.296	2.246	2.288	2.337	2.453	2.500	2.451
	Ebn at Noon		276	288	295	284	279	273	268	268	269	271	270	267
		Edh at Noon	24	30	33	40	41	41	43	40	35	28	23	22
(44) All-Sky Solar Radiation	RadAvg		612	936	1204	1518	1796	1992	2042	1744	1436	979	668	505
	RadStd		57	65	115	126	134	167	130	119	97	120	47	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page